Online Food Delivery Analysis: Data-Driven Business Insights

Customer & Order Analysis

5. Impact of discounts on profit

Margin doesn't change with discount size. Orders with any discount average margin against no discount. There's no sign that discounts make customers spend more. Discounts show no impact on profit margin, order value or the share of loss-making orders.

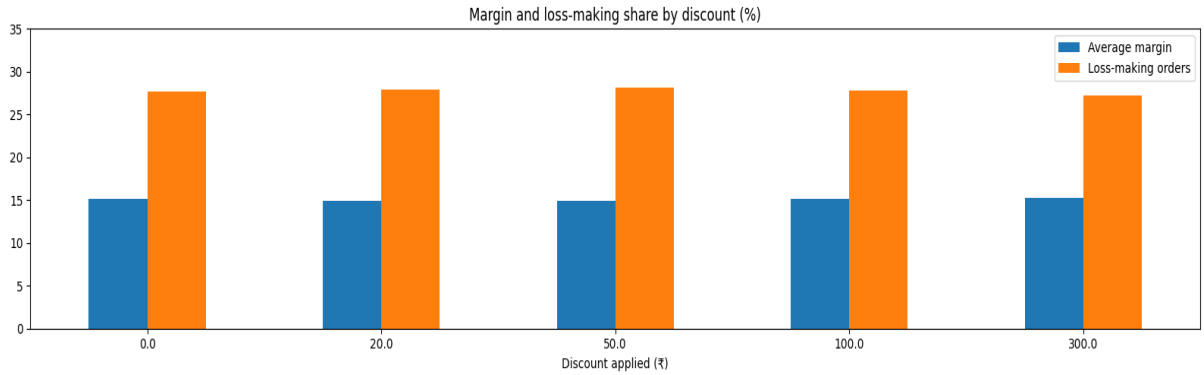
Illustrative sample records :

Discount Applied	orders	Avg margin	Loss share	avg_order_value
0.0	16650	15.1	27.7	2062.5
20.0	16569	14.9	27.9	2068.2
50.0	16690	14.9	28.1	2077.8
100.0	16776	15.1	27.8	2083.3
300.0	16600	15.3	27.2	2084.9

2 . VISUALIZATIONS

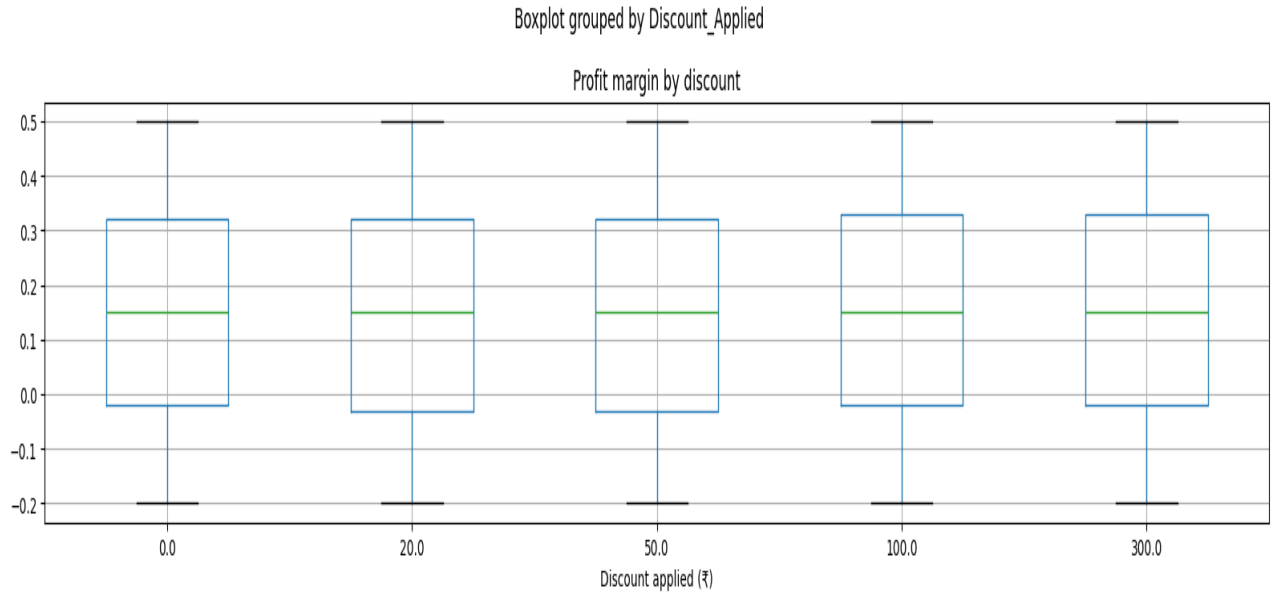
Visualization – 1

This chart shows Profit by discount amount applied.



Visualization – 2

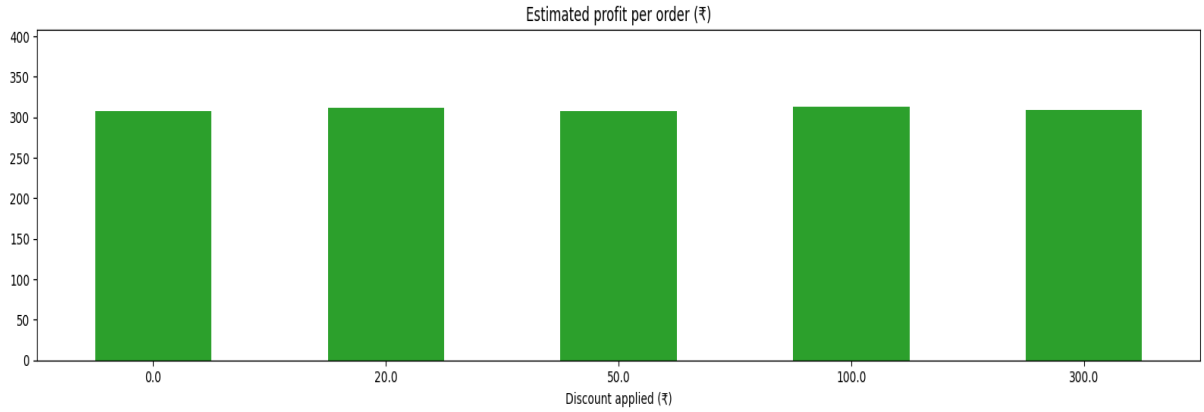
Below boxplot shows profit margin by discount.



Visualization – 3

Below bar chart shows estimated profit per order vs discount applied.

Online Food Delivery Analysis



3.ANALYTICAL INSIGHTS – DEMAND AND REVENUE

INSIGHT – 1: Discounts are everywhere.

83% of orders carry a discount, and the four discount are used about equally, each on roughly 20% orders. Discounts are a standard feature of the business, not an occasional promotion.

INSIGHT – 2 : Profit margin stays the same.

Margins sit between 14% and 15% at every discount level. Whether a customer gets nothing or ₹300 off, the margin is about 15%.

INSIGHT – 3: Discounts don't change customer behaviour.

Customers who get bigger discounts on average don't place more orders. Delivery ratings don't differ by discount and weekdays and weekends use discounts equally. Cancellations are the one borderline result, but there's no trend by size.

Here some points to fix the discount problem:

- Test smaller discounts before changing anything.
- Give discounts to the customers who need them, not for all orders.
- Set a discount budget and track it**.**

4.ANALYTICAL INSIGHTS-OPERATIONS AND CUSTOMER EXPERIENCE:

INSIGHT – 4: Calculate discount Percentage vs. Profit Margin

Compare profit margins across different discount levels, such as 5%, 10%, 20%, and 30%. Identify discount levels that attract customers while maintaining a healthy profit margin.

INSIGHT – 5 : Analyse Customer Retention Through Discounts

Target loyalty incentives toward customers likely to make additional profitable purchases.

INSIGHT – 6: Compare discount Impact Across Customer Segments

Allocate discounts to customer segments where promotions generate incremental profitable activity.